Lab 4:
CLIENT-SIDE SCRIPTING (JAVASCRIPT)

Topic	Web page development using HTML and Scripting (JavaScript)
Domain of Learning	Psychomotor (P2: Set; P3: Guided Respond; P4: Mechanism)
Learning objective	To apply HTML and JavaScript effectively to develop interactive, functional, and visually appealing web pages, demonstrating a thorough understanding of syntax, structure, event handling, and DOM manipulation to enhance user experience.
Lab activity objective	Apply HTML tags and JavaScript functions effectively to meet specific requirements, demonstrating a solid understanding of event handling, loops, and DOM manipulation in web development.

Instruction: Answer all questions.

1. Write a complete HTML document with JavaScript code that automatically changes the paragraph text from `"Welcome to my website!"` to `"Hello there!"` when the webpage loads.

   Given code:

```
<div id="myElement">
  <p>Welcome to my website!</p>
</div>
```

2. Based on the HTML code below, write a complete HTML document with a JavaScript function named checkInput() that uses **if...else statements** to validate the user's input. The function should display an alert based on the following conditions:
   a. If the number is lower than 10, display "The number is lower than 10".
   b. If the number is higher than 10, display "The number is higher than 10".
   c. If the number is equal to 10, display "The number is equal to 10".
   d. If the input is not a valid number (e.g., alphabets or empty input), display "Please insert a number!".

   Given HTML code:

```
<input type="text" id="userInput" placeholder="Enter a number">
<button onclick="checkInput()">Check</button>
```

3. Complete the JavaScript code below using a **for loop** to display numbers from 1 to 5, as shown in the expected output. Write your answer in a complete HTML document.

Given code:

```
<p id="demo"></p>


    <script>
      let text = "";

      for (let i = 0; /* Complete here */) {
        text += "The number is " /* Complete here */
    }

    document.getElementById("demo").innerHTML = text;
    </script>
```

Expected output:

```
The number is 1
The number is 2
The number is 3
The number is 4
The number is 5
```

4. Upload all three HTML files to your GitHub repository. Refer to the GitHub submission guide provided.

5. Copy your answer and screenshot the output in Microsoft Word. Include your GitHub repository link. Submit through Author in PDF format.

HOW TO SUBMIT YOUR JAVASCRIPT CODE TO GITHUB

Follow these steps after completing Questions 1, 2 and 3. You can upload your files directly through the GitHub website; Git commands are not required.

Step 1: Prepare your files

Save each complete HTML answer separately as Q1.html, Q2.html and Q3.html. Check that each file opens and works correctly in a web browser.

Step 2: Create a GitHub repository

Visit https://github.com and sign in (or create an account). Select New repository, enter BIT21503-Lab4-JavaScript as the repository name, choose Public, then click Create repository.

Step 3: Upload your files

Open the new repository. Select Add file > Upload files (or the upload link shown in an empty repository). Drag and drop Q1.html, Q2.html and Q3.html, or choose the files from your computer.

Step 4: Save your changes

Click Commit changes to save the uploaded files in the repository. Check that all three files appear in the file list.

Step 5: Copy your repository link

Copy the URL from your browser address bar, for example: https://github.com/your-username/BIT21503-Lab4-JavaScript

Step 6: Submit your lab report

In Microsoft Word, include the code for each question, screenshots of the corresponding browser outputs, and your GitHub repository link. Export the report as PDF and submit it through Author.

Before submitting: Make sure your repository is Public, the link opens correctly, and all three HTML files are included.